Saturating a Pell witness to remove a radical

Smaller explicit Diophantine equations for syntactic consistency of ZFC

Research contribution prepared for Vladimir Reshetnikov
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Abstract

We give four explicit polynomials in ten strictly positive integer unknowns, with operation/degree profiles 92/283, 93/278, 98/258, and 99/254. They improve the inspected ProveIt ten-positive baselines 99/368 and 105/340. The substantive change absorbs the first Pell ordinate into the strong Pell witness, making two required squares coprime factors of one square. An elementary congruence excludes the exceptional cancellation between the two remaining radicals. The final equation is consequently a four-factor norm instead of an eight-factor norm. Both directions of positive-witness existence are proved; the reverse direction uses a finite Pell-matrix period and the explicit auxiliary witness already available in the source construction.

The same saturation principle, with a compatible rescaling of the auxiliary coordinate and full-residue interpolation, gives an equation in twelve unrestricted integers with profile 123/1084. It improves both current twelve-integer parents. Polynomial coordinate changes with profiles 143/1200 and 146/1184 are retained as controls in an appendix. A final appendix develops an explicit degree-14 local equational proof kernel as a component of a different future source encoding.

The consistency calibration is inherited from the precisely identified ordinary native, compiler, and refutation theorems in the repository. This contribution supplies ordinary proofs formalizable in ZFC and reproducible finite certificates; it does not supply a new end-to-end Lean or Rocq proof, prove consistency, establish a global optimum, or claim a new minimum number of unrestricted integer variables.

Contents

1 Results, comparison, and the logical claim **3**

1.1 What is new in this round 4

2 The complete ten-positive polynomial **4**

2.1 The direct coordinate version 5

2.2 The exact arithmetic schedules 5

3 Why the product of two squares can replace two radicals **5**

4 An unbounded arithmetic proof of the saturation step **6**

4.1 Two roots recover the three required Pell conditions 6

4.2 The old strong witness and the early inequalities 7

4.3 Why the product center suffices on both charts . . . 9
4.4 Transport soundness without a large majorant . . . 9
4.5 Licenses for the inherited index and projection proof . . . 10
4.6 Completeness: choose a divisible Pell period . . . 11

5 Exact degrees and highest homogeneous forms 12
5.1 Literal degrees and highest homogeneous forms . . . 12

6 A saturated equation over twelve unrestricted integers 14
6.1 Complete twelve-integer polynomial . . . 15
6.2 Signed bootstrap, separation, and source restoration . . . 16
6.3 Completeness with a fixed strong witness . . . 17
6.4 Attained degree and reproducible cost . . . 18

7 Proof dependencies, reproducibility, and limitations 19
7.1 The inherited semantic boundary . . . 19
7.2 What the executable checks establish . . . 20

8 Further research questions 20

A The complete fixed coefficient recipe 21

B Coordinate-shear controls over twelve integers 23
B.1 Fixed coefficients and complete equations . . . 24
B.2 Unconditional integer zero-set bijections . . . 25
B.3 Exact degree certificates . . . 25
B.4 Cost and what the computations certify . . . 26

C An alternative source: an explicit equational proof kernel 27
C.1 A total coding of terms . . . 27
C.2 The seven-variable local polynomial . . . 28
C.3 Finite proof checking and the remaining obligations . . . 29
C.4 An explicit combined row polynomial . . . 30

1 Results, comparison, and the logical claim

The source snapshot is ProveIt commit `84e16e43092614bae262ca59651cb9d77e31a255`. All comparisons in this article refer to that snapshot. Its active research directory is more recent than its placed-manuscript catalogue: the complete displayed page has 101 operations and degree 368, while the accepted zero-offset evaluator already has 99 operations at the same degree. The direct coordinate sibling has 105 operations and degree 340. The new comparison is therefore against these later accepted evaluators, not against the older wave-ten catalogue [1, 2, 3].

Construction	Quantified domain	Binary operations	Exact degree
Pinned zero-offset parent	10 positive integers	99	368
New saturated upper center	10 positive integers	92	283
New saturated sharp center	10 positive integers	93	278
Pinned direct parent	10 positive integers	105	340
New direct upper center	10 positive integers	98	258
New direct sharp center	10 positive integers	99	254
Pinned rational scalar carrier	12 unrestricted integers	145	1200
Pinned integral scalar carrier	12 unrestricted integers	141	1456
New saturated scalar carrier	12 unrestricted integers	123	1084
Centered integral control	12 unrestricted integers	143	1200
Centered rational control	12 unrestricted integers	146	1184

An operation is one binary addition, subtraction, or multiplication, including multiplication by a numerical constant and squaring. Registers can be shared. Constants and variable inputs are leaves, not operations. Degree counts quantified variables only. This is an arithmetic-circuit measure, not bit complexity: the large fixed coefficients are reported separately and are not being made smaller.

The 92/283 form saves seven operations and 85 degrees from 99/368, approximately 7.1% and 23.1%, respectively. The 99/254 direct form uses no more operations than the former cheapest ten-positive evaluator and reduces its degree by 114, approximately 31.0%. The four new ten-positive points are different cost/degree choices. The twelve-integer saturation saves 22 operations and 116 degrees from its rational parent, and 18 operations and 372 degrees from its integral parent. They do not improve the repository's nine-positive or ten-unrestricted- integer variable counts, nor its separate low-degree quartic families.

Theorem 1.1 (Consistency interpretation and its dependency boundary)**.** *Let $N_{2,\varepsilon}$ and $N_{2,\varepsilon}^{\mathrm{dir}}$ be the completely specified polynomials below, for $\varepsilon = 0, 1$, at the coherent fixed tuple of Appendix A. The ordinary source calibration identified in Section 7, composed with the witness constructions in this article, gives*

$$\mathrm{ZFC} \vdash \left[\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{10}\, N_{2,\varepsilon}(\mathbf{x}) = 0\right] \iff \mathrm{Con}(\mathrm{ZFC}),$$

and the same statement for the direct variant. The twelve-integer form in Section 6, and the controls in Appendix B, have the analogous statement with $\mathbf{x} \in \mathbb{Z}^{12}$.

Here Con(ZFC) denotes absence of a finite syntactic refutation in the same effective ZFC calculus used by the pinned source. The displayed claim is an ordinary mathematical derivability claim, using the existing ordinary calibration, formalizable in ZFC. The archive does not contain an independent formal verification of that entire inherited theorem. Its new machine-executed checks concern the literal polynomial specifications, their algebraic identities, and their finite certificates. No assertion that either side of the equivalence is true is needed for the construction.

1.1 What is new in this round

The transport expansion $WB/(q-1) = LB + B/(q-1)$ and the general ideas of centering and saturating Pell witnesses already occur in the source collection. They are credited as such. The principal new result is the complete application of coprime-product saturation to the current ten-positive kernel: a two-root recovery proof, an inexpensive upper center, full positive completeness after changing the strong witness, and the four literal equations above. A focused search of the downloaded incoming submissions and the inspected active proofs found no earlier instance of these complete equations. This is a repository-relative comparison, not a claim of worldwide priority for coprime-square splitting. The unrestricted-integer extension applies the same idea to the signed scalar carrier. Its additional contribution is the compatible auxiliary rescaling and residue interpolation needed to preserve completeness.

2 The complete ten-positive polynomial

Choose once and for all $\varepsilon \in \{0, 1\}$. It selects a polynomial; it is not an eleventh unknown. The ten unknowns are

$$J, F, \alpha, i, T, w, Y, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The four fixed integers β, K_0, M_C, M_F are specified completely in Appendix A and as literal payloads in the archive. They satisfy

$$\beta \geq 31 \text{ odd}, \qquad K_0 > 0, \qquad 0 < M_C < \beta < M_F < 2\beta.$$

All symbols introduced next are polynomial abbreviations; none adds an existential variable. Set

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq, \\ \mu &= q^2 + 1, & k &= \eta + \zeta, & c &= kY + \eta, \qquad (1) \\ A &= \mu Y(X+1) + 1, & H &= 4A - 5, & \Delta &= A^2 - 1, \\ V_0 &= X\mu^2 Y^2, & E &= 1 + V_0(V_0+1)k^2, & U &= ic^2. \qquad (2) \end{aligned}$$

The rank and transport data are

$$\begin{aligned} R &= [(q-1)(q-F) + \alpha](q^2-1) + (M_C + qM_F)J, \\ e &= k - R - 1, & B_t &= (K_0 + w)C - F, \\ T' &= T + \alpha, & b &= cT' - R, & L &= kT'. \qquad (3) \end{aligned}$$

The saturated strong and auxiliary quantities are

$$\begin{aligned} D_f &= 1 + \Delta U^2 E, & Q &= \Delta^2 U^2 E, & \delta &= Q - 1, \\ V &= D_f b - c, & D_y &= QV^2 - 1, & D_g &= ED_f, \\ C_g &= (A + \varepsilon)UE, & K_+ &= X(3c - 2X) + 2. \qquad (4) \end{aligned}$$

Define the integer polynomials

$$\begin{aligned} P_1 &= q^3(q-1), & M &= P_1\mu^2 H, \\ Z &= P_1\mu^2\{H(n + V + C_g - LB_t) + K_+\} \\ &\qquad + H\{\mu^2[(q-1)Y - q^3 B_t] - P_1 e\}, \qquad (5) \\ f_s &= \delta Z^2, & f_g &= \delta M^2 D_g, & f_y &= M^2 D_y. \end{aligned}$$

The promised single equation is

$$\boxed{N_{2,\varepsilon} = (f_s - f_g - f_y)^2 - 4f_g f_y = 0.} \tag{6}$$

There is no division, radical, variable exponent, unbounded product, or undetermined program parameter in this polynomial. Large powers used in the fixed coefficient recipe define constants before the unknowns are introduced. The names `saturated_upper.json` and `saturated_sharp.json` correspond to $\varepsilon = 1$ and 0.

2.1 The direct coordinate version

Replace the unknown w by a positive unknown x and set $X = q + x$. In (3) replace B_t by

$$B = (qK_0 + X)C - qF.$$

Use $P_1 = q^4(q-1)$ and $M = P_1\mu^2 H$. Replace only the numerator by

$$\begin{aligned} Z_{\mathrm{dir}} = {} & P_1\mu^2\{H(n + V + C_g - LB) + K_+\} \\ & + H\{\mu^2[q(q-1)Y - q^3B - (q-1)X] - P_1 e\}. \end{aligned} \tag{7}$$

All other formulas, including the final norm, stay as displayed. In particular the direct P_1 contains q^4, not q^3. Its normalized center has the additional term $-X/q^4$; this is a proof identity, not a division in the evaluator. The retained direct-coordinate argument recovers $q \mid X$ only after the whole-product projection has removed the H denominator. No earlier cancellation of q is used.

2.2 The exact arithmetic schedules

The JSON files list every gate in topological order as `[register, operation, operand1, operand2]`. All gates, inputs, and fixed ports are live. For efficient strong arithmetic, put

$$U_E = UE, \quad \Delta = (A-1)(A+1), \quad S_E = \Delta U_E, \quad B_f = S_E U, \quad D_f = B_f + 1, \quad Q = \Delta B_f.$$

For the upper center $(A+1)U_E$, the literal $A+1$ is already needed in Δ. The sharper AU_E requires one extra addition to obtain A. The final norm takes thirteen gates from M^2 through its output, including the product $D_g = ED_f$. The operation breakdowns are

Variant	Multiplications	Additions	Subtractions	Total
Upper	55	20	17	92
Sharp	55	21	17	93
Direct upper	59	21	18	98
Direct sharp	59	22	18	99

The distributed transport numerator used here was already available in the earlier dyadic submission. The new schedules compose it with the saturated kernel, not with a re-labelled old finalizer.

3 Why the product of two squares can replace two radicals

The central design choice is the congruence

$$D_f = 1 + \Delta U^2 E \equiv 1 \pmod{E}.$$

Thus E and D_f are coprime. Once the two-root norm forces ED_f to be a square, both factors must individually be squares:

$$E = \tau^2, \qquad D_f = f^2.$$

The proof-only integer $I = U\tau$ then satisfies the original strong equation

$$f^2 - \Delta I^2 = 1, \qquad c^2 \mid I.$$

There is no additional unknown for I or τ in the new equation.

One must also exclude a false zero caused by cancellation of two equal irrational roots. Here that exclusion is built into the same factor E: if

$$ED_f = \frac{QV^2 - 1}{Q - 1},$$

then $(Q-1)ED_f = QV^2 - 1$, whereas $E \mid Q$ makes the two sides congruent to 0 and -1 modulo $E > 1$. This contradiction is the reason the two-root finalizer is sound. No unsupported assertion that a rational sum of radicals makes every radical rational is used.

The converse is also substantive. An old strong ordinate need not be divisible by τ. Instead, after constructing the canonical outer witnesses, choose a positive Pell-matrix period modulo $c^2\tau$. The resulting new ordinate is divisible by $c^2\tau$, so $U = I/\tau = ic^2$ for a positive integer i. The explicit auxiliary Pell solution is large enough to make the final n positive despite the new upper center. These details are proved in the next section.

4 An unbounded arithmetic proof of the saturation step

We prove the equivalence for both charts and for either of the two fixed centers $C_g = (A + \lambda)UE$, where $\lambda = \varepsilon \in \{0, 1\}$. Write $K_{\mathrm{pos}} = K_+$ and $G = D_g = ED_f$. The other notation is that of the complete equation above. Throughout this section, all ten independent coordinates are strictly positive integers. A "root" denotes a positive real square root only inside the proof; the polynomial evaluator has no root operation.

The inherited input is the accepted ten-positive zero-offset construction, together with its coherent native/controller/refutation calibration. Its proof imports the universal Pell facts and the source61 lemmas named `lem:rank`, `lem:index`, and `lem:auxcomplete`. We state their interfaces at the points where they are used. The old strong witness is not retained literally: the proof will recover $I = U\sqrt{E}$ and, in the converse direction, choose a new strong Pell period. The complete original port tuple and the complete BFS tuple are separate allowed calibrations; individual ports are never interchanged.

4.1 Two roots recover the three required Pell conditions

Lemma 4.1 (Root recovery by a coprime product)**.** *At every positive integer zero of the displayed polynomial, there are positive integers τ, f, y such that*

$$E = \tau^2, \qquad D_f = f^2, \qquad QV^2 - (Q-1)y^2 = 1, \tag{8}$$

and

$$\frac{Z}{M} = \varepsilon_g f\tau + \varepsilon_y y, \qquad \varepsilon_g, \varepsilon_y \in \{-1, 1\}. \tag{9}$$

No sign restriction on Z or V is needed for this conclusion.

Proof. Before using the equation, $E > 1$, $D_f > c + 1$, $Q > 1$, $M > 0$, and $\delta = Q - 1 > 0$. Since $V = D_f b - c$ and b is an integer, either $b = 0$ and $V = -c$, or $|V| \geq D_f - c > 1$. In particular $D_y > 0$. Set

$$G = ED_f, \qquad Y_* = \frac{D_y}{\delta} = \frac{QV^2 - 1}{Q - 1}.$$

The two positive rational numbers G, Y_* cannot coincide. Indeed, $G = Y_*$ would imply

$$Q(V^2 - G) = 1 - G.$$

Both Q and G are divisible by E, so this would give $E \mid 1$. This simple obstruction replaces a radical size or sign separation.

At a zero put $z = Z/M$. The equation is exactly

$$(z^2 - G - Y_*)^2 = 4GY_*.$$

If $z = 0$, then $G = Y_*$, which was excluded. For $z \neq 0$,

$$\left(\frac{z^2 + G - Y_*}{2z}\right)^2 = G.$$

The complementary rational number has square Y_*. Thus both positive roots are rational. Since G is an integer, it is an integer square. Moreover

$$D_f = 1 + \Delta U^2 E \quad \Longrightarrow \quad \gcd(E, D_f) = 1.$$

A square product of positive coprime integers has square factors. Write $E = \tau^2$, $D_f = f^2$, and $g = f\tau = \sqrt{G}$.

For clarity, rational integrality of the remaining adjacent-Pell root has the following elementary descent. If $y = a/m > 0$ in lowest terms satisfies $Qv^2 - (Q-1)y^2 = 1$ with integer $v > 0$, then $m^2 \mid Q - 1$. For $v > 1$ use

$$v' = (2Q - 1)v - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2Qv.$$

The first coordinate is integral, the same norm is preserved, and $v < y < Qv/(Q-1)$ gives $-v < v' < v$. The new first coordinate cannot vanish. After taking absolute values, its positive integer size has decreased. The reduced denominator of y' is still m: $2Q - 1 \equiv 1 \pmod{m}$ and a is coprime to m. Descent reaches $v = 1$, where $y = 1$, so $m = 1$. Apply this argument initially to $v = |V|$. The signed factorization now gives (9). □

4.2 The old strong witness and the early inequalities

Define only inside the proof

$$I = U\tau. \tag{10}$$

Then I is a positive integer, $c^2 \mid I$, and

$$f^2 - \Delta I^2 = 1, \qquad Q = (\Delta I)^2, \qquad V = f^2(cT' - R) - c. \tag{11}$$

These are exactly the normalized strong and auxiliary identities of the accepted zero-offset proof. Its old strong input could be $i_{\text{old}} = I/c^2 = i\tau$; it is not the new literal coordinate i.

Lemma 4.2 (Folding and size before transport)**.** *At every positive zero there are indices $p, n_0 \geq 1$, with p odd, such that*

$$c = \psi_A(p), \qquad k = 2\psi_{2V_0+1}(n_0), \qquad n_0 \geq 2.$$

Consequently

$$k \geq 8V_0 + 4, \qquad c > kY \geq k > 8q^5, \qquad K_{\mathrm{pos}} > 0, \qquad H > 3X > 0. \tag{12}$$

These conclusions do not use the sign of the transport numerator.

Proof. The inherited strong/auxiliary folding lemma is uniform for $A \geq 3$. Its hypotheses are the strong norm, $c \leq I$, $c^2 \mid I$, the adjacent auxiliary norm, and $V \equiv -c \pmod f$. They follow from (8)–(11). It gives $c = \psi_A(p)$ for odd $p \geq 1$. The unchanged first norm gives the stated expression for k. If $n_0 = 1$, then $k = 2$; positivity forces $\eta = \zeta = 1$, and $c = 2Y + 1$ lies strictly between $\psi_A(1) = 1$ and $\psi_A(2) = 2A$. This is impossible. Thus $k \geq 2\psi_{2V_0+1}(2) = 8V_0 + 4$. The inequalities in (12) now use $V_0 = X\mu^2Y^2$, $X \geq q$, and $\mu = q^2 + 1$. In particular $c > 2X$, so $K_{\mathrm{pos}} = X(3c - 2X) + 2 > 0$; the formula $H = 4\mu Y(X + 1) - 1$ gives $H > 3X$. □

The port bounds of the native mask give, for every positive assignment,

$$\begin{aligned} R &< q^4 + (\alpha + 2)q^2, \\ |R| &< q^4 + Fq^3 + (\alpha + 2)q^2, \\ \frac{|R|}{\mu^2} &< F + \alpha + 1. \end{aligned} \tag{13}$$

For the last inequality, multiplying by $\mu^2 = (q^2 + 1)^2$ leaves the positive difference

$$F(q^4 - q^3 + 2q^2 + 1) + \alpha(q^4 + q^2 + 1) + 1.$$

The first bound and $c > 8q^5$ show that $b = c(T + \alpha) - R > 0$, hence $b \geq 1$ and $V = D_f b - c > 1$. Put

$$u = y - V > 0, \qquad \rho = \frac{e}{\mu^2}.$$

The adjacent norm and $\Delta D_f = Q + \Delta < 2Q - 2$ give

$$u = \frac{V^2 - 1}{(Q - 1)(y + V)} < \frac{V}{2(Q - 1)} < \frac{b}{\Delta}.$$

Using $c < 2kY$, $\Delta > 8Y$, $\Delta > \mu^2$, and (13), we obtain

$$u < \frac{kT'}{4} + F + \alpha + 1, \qquad \rho < \frac{k}{\mu^2} + F + \alpha + 1. \tag{14}$$

If $C = q - F - \alpha \geq 1$, then $q(q - F) - C = q\alpha + (q - 1)C > 0$ and therefore $R > 0$. In that case the sharper inequalities are

$$u < \frac{kT'}{4}, \qquad \rho < \frac{k}{\mu^2}. \tag{15}$$

At this stage ρ may still be rational. No denominator has yet been cancelled.

4.3 Why the product center suffices on both charts

Lemma 4.3 (Upper and sharp centers)**.** *At every positive zero,*

$$g < C_g, \qquad 0 < C_g - g < (\lambda + 1)UE \le 2UE. \tag{16}$$

For the sharp center $C_0 = AUE$, the stronger estimate

$$C_0 - g > \frac{UE}{4A} > X \tag{17}$$

also holds. In particular, for either center and either root sign, $\varepsilon_g g - C_g + X/q^4 < 0$.

Proof. Since $I > 1$ and $f^2 = A^2I^2 - I^2 + 1$, $(A-1)I < f < AI$. Multiplication by τ proves (16). For the sharper lower bound use

$$C_0^2 - G = U^2E^2 - E.$$

Here $U^2E > 2$ and $g < C_0$, so $(C_0 - g) > UE/(4A)$. The next estimate is unconditional on the positive coordinate domain: $k \ge 2$, $c > kY$, $U \ge c^2$, and $E > V_0^2k^2$ imply

$$UE > V_0^2k^4Y^2 \ge 16V_0^2Y^2.$$

Also $A = \mu Y(X+1) + 1 < 4\mu XY$. Hence

$$\frac{UE}{4A} > \frac{V_0^2Y}{\mu X} = X\mu^3Y^5 > X.$$

In particular the direct estimate uses neither $q \mid X$ nor any decoded source information. □

4.4 Transport soundness without a large majorant

On the ordinary chart write $W = 1 + (q-1)L$ and

$$t = \frac{B_t}{q-1}, \qquad s = \frac{Y}{q^3} > 0, \qquad \kappa = \frac{K_{\mathrm{pos}}}{H} > 0.$$

The exact normalized carrier and (9) give

$$n = (\varepsilon_g g - C_g) + (\varepsilon_y y - V) - \kappa - s + Wt + \rho.$$

By Lemma 4.3, every zero satisfies

$$n < u + Wt + \rho. \tag{18}$$

Suppose $B_t \le -1$. If $C \ge 1$, then $Wt < -L = -kT'$; (15) gives

$$n < \frac{kT'}{4} + \frac{k}{\mu^2} - kT' < 0.$$

If $C \le 0$, then $B_t = (K_0 + w)C - F \le -F$ and $Wt < -FL$. By (14),

$$n < \frac{kT'}{4} + \frac{k}{\mu^2} + 2F + 2\alpha + 2 - FkT' < 0.$$

For completeness, the negative of the expression on the right is increasing in $F, T, \alpha \ge 1$ when $k \ge 8$; its minimum is $3k/2 - k/\mu^2 - 6 > 0$. Both cases contradict $n > 0$. We have proved $B_t \ge 0$, and then $(K_0 + w)C \ge F > 0$ gives $C > 0$.

On the direct chart put $W = 1 + q(q-1)L$. The normalized carrier differs by the transport term $-WB/[q(q-1)]$ and the additional term $-X/q^4$. Lemma 4.3 absorbs the resulting $+X/q^4$ in the equation for n. Thus the same upper bound holds with Wt replaced by $WB/[q(q-1)]$. If $B \le -1$ and $C \ge 1$, this term is smaller than $-L$. If $C \le 0$, then $B = (qK_0 + X)C - qF \le -qF$, so it is smaller than $-qFL \le -FL$. The identical two contradictions give $B \ge 0$ and $C > 0$. Neither argument assumes the divisibility $q \mid X$.

4.5 Licenses for the inherited index and projection proof

We record the order of implications because it prevents circular cancellation arguments. Once transport is nonnegative, the native mask estimate gives $3q < R < q^4 - 2$. The recovered strong witness satisfies $c^2 \mid I$ and $c > 2R$. Since $c = \psi_A(p) > 2$ and $A \ge 3$, one also has $c > 2p$.

The source61 rank lemma says that $c = \psi_A(p) > 2$, $f^2 - \Delta I^2 = 1$, $f, I > 0$, and $c^2 \mid I$ imply

$$f = \chi_A(m), \quad I = \psi_A(m), \quad pc \mid m, \quad c \mid m, \quad m > 2p.$$

Put $S = \Delta I$. Then $S \ge 2$, $c \mid S$, and $S^2 = \Delta(f^2 - 1)$. The two required auxiliary congruences are

$$V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c};$$

the second uses $f^2 \equiv 1 \pmod{c}$. The source61 index lemma, with $0 < p, R < c/2$, now gives $p = R$.

The index R is odd. If q were odd, then J would be even and the native mask formula would make R even. Thus q is even and J is odd. Only now may one use

$$\gcd(\mu^2, q^3(q-1)H) = 1 \quad \text{or, on the direct chart,} \quad \gcd(\mu^2, q^4(q-1)H) = 1.$$

Indeed $H \equiv -1 \pmod{\mu}$, and even q makes $\gcd(q-1, \mu) = 1$. The signed carrier is integral, so the unchanged denominator argument gives $\mu^2 \mid e$.

All following uses of the zero-offset parent are now licensed: its first-index argument gives $n_0 = (R+1)/2$; its offset-free ratio estimate gives the same half-binomial interval and size bound; and its whole-product projection gives $X = 2^R$. More explicitly, on the ordinary chart the integral carrier yields $H \mid q^3(q-1)K_{\mathrm{pos}}$. Writing

$$K_{\mathrm{quad}} = X\big(X + 2(A-2)c\big) - 1, \qquad K_{\mathrm{pos}} = HXc - 2K_{\mathrm{quad}},$$

one cancels only 2 (since H is odd). The universal projection therefore yields

$$H \mid q^3(q-1)(X - 2^R)(2^R X + 1).$$

The entire product has absolute value less than H by the retained ratio and height bounds; hence $X = 2^R$. One does not cancel an unverified unit modulo H. The direct parent supplies exactly the corresponding product argument with its extra carrier factor.

The new center causes no change to these interfaces. Both C_g and $g = f\tau$ are integers, so the only nonintegral terms remain the old fractions. On the direct chart, after the main projection makes K_{pos}/H integral, the residual relation is

$$\frac{Y}{q^3} - \frac{WB}{q(q-1)} - \frac{X}{q^4} \in \mathbb{Z}.$$

Its numerator modulo q gives $q \mid X$. Since $X = 2^R$, q is a power of two; $R > 3q$ then gives $q^4 \mid X$. Set $w = X/q$ and use $B = qB_t$ to finish the old source decoding.

The remaining parent steps give $R \equiv 3 \pmod 4$, the exact half-binomial value of Y, and all native source divisibilities. Its final reconstruction uses a *fresh unscaled* source61 parameter $A_{\text{source}} = Y(X+1)+2$. It neither divides the new strong witness nor identifies this unscaled parameter with A. The fixed native/controller calibration then produces a finite syntactic ZFC refutation. This proves soundness for each new center and chart.

4.6 Completeness: choose a divisible Pell period

Suppose a finite ZFC refutation exists. The parent's canonical construction gives the outer and first/main data, with

$$E = \tau^2, \quad c = \psi_A(R), \quad R \equiv 3 \pmod 4, \quad R > 3q,$$

and canonical integers $t \geq 0$, $\rho \geq 0$, $s > 0$, and $\kappa > 0$. Choose the strong pair anew rather than reusing the parent's particular index.

Let $N_0 = c^2\tau$. The determinant-one matrix

$$M_A = \begin{pmatrix} A & \Delta \\ 1 & A \end{pmatrix}$$

has finite positive order in the group of invertible matrices over $\mathbb{Z}/N_0\mathbb{Z}$. At a positive multiple m of that order,

$$I = \psi_A(m) > 0, \qquad f = \chi_A(m) > 0, \qquad N_0 \mid I, \qquad f^2 - \Delta I^2 = 1.$$

Set $i = I/(c^2\tau) > 0$. Then the literal $U = ic^2$ satisfies $U = I/\tau$, and the new D_f, Q are exactly $f^2, (\Delta I)^2$. This construction requires divisibility of a Pell ordinate by a prescribed integer, which a finite period supplies. It requires no intersection theorem for two unrelated Pell sequences.

The source61 auxiliary-completeness lemma assumes only $A \geq 2$, $R \geq 3$, $R \equiv 3 \pmod 4$, $c = \psi_A(R)$, $f^2 - \Delta I^2 = 1$, $f, I > 0$, and $c \mid I$. All hold for this new pair. In particular the lemma fixes neither the strong index nor the parity of I/c^2. Its explicit witness, with $S = \Delta I$, $Q = S^2$, and $R = 2v+1$, is

$$\begin{aligned} V_* &= \chi_S(R)/S = Q_v(S^2), \qquad y = \psi_S(R), \\ T_* &= \frac{V_* + c + Rf^2}{cf^2}, \qquad QV_*^2 - (Q-1)y^2 = 1. \end{aligned}$$

The lemma makes T_* a positive integer. Its usual Pell growth bound gives, since canonical $v \geq 2$,

$$V_* \geq (4Q-3)^v \geq (3Q)^v \geq 9Q^2.$$

Because $c < S$, $f^2 < Q$, and $\alpha < q < S$,

$$T_* > \frac{V_*}{cf^2} > \frac{V_*}{S^3} \geq 9S > \alpha.$$

Thus $T = T_* - \alpha$ is positive and the literal auxiliary register equals V_*. For $u = y - V_* > 0$, using $V_* \geq 2$ and $y < 2V_*$,

$$u = \frac{V_*^2 - 1}{(Q-1)(y+V_*)} > \frac{V_*}{4Q} \geq \frac{9}{4}S^2 = \frac{9}{4}\Delta^2 I^2. \tag{19}$$

Here $U = ic^2 > 1$, so $UE = I\tau < I^2$. Also $\kappa < c < I$ and $s < Y < c < I$; the first bound follows from $K_{\text{pos}} < 3Xc$ and $H > 3X$. Define the integer

$$n = (g - C_g) + (y - V_*) - \kappa - s + Wt + \rho.$$

Its total negative loss is less than

$$2UE + \kappa + s < 2I^2 + 2I < 4I^2,$$

whereas (19) exceeds $4I^2$, since $\Delta \geq 3$. Thus $n > 0$. The normalized center equals $g + y$, and one factor of the two-root norm vanishes. This is completeness on the ordinary chart for both centers. The explicit auxiliary witness provides the needed growth; no unproved cofinality assertion has been inserted.

For the direct chart, transfer this newly constructed ordinary zero. It has $X = 2^R > q$, $q^4 \mid X$, and $t = B_t/(q-1) \geq 0$. Set

$$\begin{aligned} x &= X - q, \\ W_{\mathrm{ord}} &= 1 + (q-1)L, \qquad W_{\mathrm{dir}} = 1 + q(q-1)L, \\ n_{\mathrm{dir}} &= n_{\mathrm{ord}} + (W_{\mathrm{dir}} - W_{\mathrm{ord}})t + X/q^4. \end{aligned}$$

Every coordinate is positive. All kernels and C_g agree, and the normalized centers agree. In fact $M_{\mathrm{dir}} = qM_{\mathrm{ord}}$ and $Z_{\mathrm{dir}} = qZ_{\mathrm{ord}}$, so the new norm scales by q^4 and remains zero.

Theorem 4.4 (Semantic equivalence of the four saturated equations)**.** *For either fixed center and either chart, with any one complete coherent calibrated port tuple, the constructed polynomial has a zero in ten strictly positive integers if and only if there is a finite syntactic ZFC refutation. Consequently ZFC proves that its lack of positive integer solutions is equivalent to* Con(ZFC)*.*

Proof. Soundness and completeness were proved above, using the same arithmetizable source calibration as the parent. Negating the resulting existence equivalence gives the asserted consistency equivalence. The argument is ordinary mathematical proof formalizable in ZFC; it is not an assertion that a new Lean, Rocq, or encoded internal-ZFC proof artifact has been produced. □

5 Exact degrees and highest homogeneous forms

This section proves the global degree bounds and their attainment for the four literal circuits. Write $\gamma = \varepsilon$ for the fixed center choice; it is not another unknown. The square-recovery argument and the complete witness constructions were proved in the preceding section.

5.1 Literal degrees and highest homogeneous forms

Set $q_0 = \beta J$ and $k_0 = \eta + \zeta$. With the port values fixed as constants, the four exact results are

circuit	live binary operations	total degree
$X = wq,\ C_1 = (A+1)UE$	92	283
$X = wq,\ C_0 = AUE$	93	278
$X = q + x,\ C_1 = (A+1)UE$	98	258
$X = q + x,\ C_0 = AUE$	99	254.

The operation counts refer to the supplied literal graphs after discarding unused nodes; they charge additions, subtractions and multiplications once each, including multiplication by a fixed integer.

Exact-degree certificate. For the $X = wq$ circuits, the highest homogeneous forms are

$$[N_{2,1}]_{283} = 2^{11} q_0^{149} w^{29} Y^{61} i^{10} k_0^{32} (q_0 - F)^2,$$
$$[N_{2,0}]_{278} = 2^{10} q_0^{146} w^{28} Y^{60} i^{10} k_0^{32} (q_0 - F)^2.$$

For the direct circuits, put $x_0 = q_0 + x$. The highest forms are

$$[N_{2,1}^{\mathrm{dir}}]_{258} = 2^{11} q_0^{124} x_0^{29} Y^{61} i^{10} k_0^{32} (q_0 - F)^2,$$
$$[N_{2,0}^{\mathrm{dir}}]_{254} = 2^{10} q_0^{122} x_0^{28} Y^{60} i^{10} k_0^{32} (q_0 - F)^2.$$

These are nonzero polynomials for every fixed $\beta > 0$. Therefore they certify exact degrees for the actual coefficient tuple.

Proof. Scale all ten witnesses by t, keep the coefficient ports fixed, and work in the formal Laurent-series field at $t = \infty$ over the rational function field in the ten unscaled witnesses. A series degree is the largest exponent of t. For polynomials it is their ordinary total degree. Choosing formal square roots with the indicated square leading forms is justified by the binomial series; each relative correction has negative degree. No analytic convergence is used.

In either source the literal numerator satisfies

$$Z := Z_{\mathrm{num}}/M = V + C_\gamma + \mathcal{E},$$

where the rational remainder has degree at most 4. For $X = wq$, it is

$$\mathcal{E} = n - LB_t + K_{\mathrm{pos}}/H + Y/q^3 - B_t/(q-1) - e/\mu^2, \quad L = k(T+\alpha), \quad H = 4A - 5,$$

and $M = q^3(q-1)\mu^2 H$. For $X = q + x$, replace the remainder by

$$\mathcal{E} = n - LB + K_{\mathrm{pos}}/H + Y/q^3 - B/[q(q-1)] - X/q^4 - e/\mu^2,$$

and retain the extra factor of q in $M = q^4(q-1)\mu^2 H$. Both denominators have degree 13. These rational identities are exact readings of the respective polynomial graphs, not additional defining equations.

Choose formal roots $g^2 = ED_f$ near AUE and $y^2 = D_y/\delta$ near V. The two exact identities are

$$(AUE)^2 - ED_f = U^2E^2 - E, \tag{D1}$$

and

$$(y - V)(y + V) = (V^2 - 1)/\delta. \tag{D2}$$

The degrees controlling the argument are

	A	U	E	D_f	δ	V	M	g	$C_1 - g$	$C_0 - g$
$X = wq$	5	5	18	38	48	42	13	28	23	18
$X = q + x$	4	5	16	34	42	38	13	25	21	17.

Equation (D2) gives $\deg(y - V) = -6$ and -4, respectively. Equation (D1) gives

$$C_0 - g = \frac{U^2E^2 - E}{AUE + g}, \qquad C_1 - g = UE + (C_0 - g).$$

The leading forms of these two defects are $u_0 e_0/(2a_0)$ and $u_0 e_0$, respectively, where the subscripts denote leading forms. In every case the defect degree is greater than 4, so it dominates both $\mathcal{E}$ and $V - y$.

The exact four-factor norm identity is

$$N_2 = \delta^2 M^4 \prod_{\sigma,\rho\in\{-1,1\}} (Z + \sigma y + \rho g). \tag{D3}$$

The two factors with $\sigma = 1$ have leading form $2v_0$. The factor with $\sigma = -1, \rho = 1$ has leading form $2g_0$. The last factor has leading form u_0e_0 for C_1 and $u_0e_0/(2a_0)$ for C_0. Thus the four degree totals are exactly

$$\begin{aligned}
2(48) + 4(13) + 2(42) + 28 + 23 &= 283,\\
2(48) + 4(13) + 2(42) + 28 + 18 &= 278,\\
2(42) + 4(13) + 2(38) + 25 + 21 &= 258,\\
2(42) + 4(13) + 2(38) + 25 + 17 &= 254.
\end{aligned}$$

To verify the coefficients without any large-numeral expansion, for $X = wq$ put

$$a_0 = q_0^3 wY, \quad u_0 = ik_0^2 Y^2, \quad e_0 = q_0^{10} w^2 Y^4 k_0^2, \quad r_0 = q_0^3(q_0 - F).$$

Then

$$d_0 = a_0^4 u_0^2 e_0, \quad m_0 = 4q_0^8 a_0, \quad v_0 = -a_0^2 u_0^2 e_0 r_0, \quad g_0 = a_0 u_0 e_0.$$

Multiplying the leading factors in (D3) gives

$$8d_0^2 m_0^4 v_0^2 g_0 u_0 e_0 \quad \text{or} \quad 4d_0^2 m_0^4 v_0^2 u_0^2 e_0^2,$$

respectively. These simplify to the two asserted $X = wq$ forms. For the direct source replace

$$a_0 = q_0^2 x_0 Y, \qquad e_0 = q_0^8 x_0^2 Y^4 k_0^2, \qquad m_0 = 4q_0^9 a_0;$$

the remaining abbreviations retain their formulas. The same products give the two direct forms. Every displayed final top form is a nonzero polynomial, proving attainment as well as the upper bound. □

Independent exact polynomial replay. The standard-library verifier evaluates every gate of each supplied graph as a univariate polynomial over the integers. On the ray where all ten witnesses equal t and $(\beta, K_0, M_C, M_F) = (2, 3, 5, 7)$, it obtains, in table order,

$$2^{192}t^{283}, \qquad 2^{188}t^{278}, \qquad 2^{167}3^{29}t^{258}, \qquad 2^{164}3^{28}t^{254}$$

as the exact leading terms. This is finite arithmetic verification of one specialization. The global Laurent-series proof, including its explicit nonzero top homogeneous forms, is what establishes the claims for the real fixed coefficient tuple.

6 A saturated equation over twelve unrestricted integers

The polynomial in `integer12_saturated/rational.json` has twelve unrestricted integer unknowns, 123 binary arithmetic operations, and exact total degree 1084. It strictly improves both operation count and degree over the pinned scalar-carrier parents, which have metrics 145/1200 and 141/1456 [5]. The construction combines coprime Pell saturation with the scalar intersective carrier. Its completeness requires a rescaling of the auxiliary coordinate and the full arbitrary-residue interpolation theorem. The new rescaling and its compatibility with that retained theorem are proved below.

6.1 Complete twelve-integer polynomial

The independent coordinates are

$$(j, C, z, h, r, v, w, T, c, n, t, s) \in \mathbb{Z}^{12}.$$

Use the exact source34 `mixed-parity-unrounded-1064921` coefficient tuple, separately from the compact44 tuple of the positive-domain forms:

$$\beta = 2^{1064921} - 1, \qquad \kappa_1 = 2^{1113}, \qquad \kappa_2 = 2^{666104}.$$

The remaining fixed integers K_0, M_C, M_F are fully specified by the included six payloads and the finite source34 recipe, complete tile table, and exact machine seed. The supplied M_F is already shifted. Every coefficient is reconstructed and compared byte for byte by the checker. No free coefficient parameter remains.

Define the following ordinary polynomial abbreviations; none is another unknown. In this section E denotes the first Pell radicand.

$$\begin{aligned}
J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4,\\
X &= qW, & Y &= (2v + 1)q^3, & a &= (X + 1)Y,\\
H &= 4a + 3, & \Delta &= a^2 + H, & F &= (K_0 + W)C - z(q - 1).
\end{aligned}$$

$$\begin{aligned}
R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J,\\
k &= R + 1 + hXY, \qquad \eta = c - Yk, \qquad \zeta = k - \eta,\\
G &= (\kappa_1 F)(\kappa_2 C) - (q - \kappa_1 F - \kappa_2 C)^2,\\
\Pi &= G\eta\zeta - (t^2 - G - \eta\zeta + 1)^2, \qquad D_g = (3\Pi - 4)s^2 + 1,\\
\ell &= XY^2, \qquad E = 1 + \ell(\ell + 1)k^2, \qquad U = src^2,\\
D_f &= 1 + \Delta U^2 E, \qquad Q = \Delta^2 U^2 E, \qquad \delta = Q - 1,\\
b &= c - R + 4UET, \qquad V = D_f b - c, \qquad C_y = QV^2 - 1,\\
K &= X(X + 2ac) - 1, \qquad S = UH^5, \qquad z_c = HT + K,\\
A_c &= (z_c^2 + 3H^2)(z_c^3 - 19H^3), \qquad Z = S(n + V) - A_c.
\end{aligned}$$

Put

$$\begin{aligned}
f_s &= \delta Z^2, & f_a &= 9\delta S^2 E D_f,\\
f_b &= 81\delta S^2 D_g, & f_c &= S^2 C_y,\\
P_e &= (f_s + f_c - f_a - f_b)^2 - 4(f_s f_c + f_a f_b).
\end{aligned}$$

The entire requested equation over $\mathbb{Z}^{12}$ is

$$\boxed{P = P_e^2 - 64 f_s f_c f_a f_b = 0.} \tag{20}$$

Every operation is polynomial arithmetic. The literal circuit evaluates $81D_g$ directly as $(243\Pi - 324)s^2 + 81$, saving one multiplication. Fixed powers in the display are implemented by counted binary multiplications.

For interpretation, let $\mathcal{F}_8(Z; A, B, C)$ be the product of $Z - \epsilon_1\sqrt{A} - \epsilon_2\sqrt{B} - \epsilon_3\sqrt{C}$ over the eight sign choices. Sign invariance makes this an integer polynomial. Then

$$P = \delta^4 \mathcal{F}_8\left(Z; 9S^2ED_f, 81S^2D_g, \frac{S^2C_y}{\delta}\right). \tag{21}$$

This is only a norm interpretation of the division-free definition (20). The denominator is cleared symbolically, and δ will be shown never to vanish on integer inputs.

6.2 Signed bootstrap, separation, and source restoration

Theorem 6.1 (Twelve-integer saturated carrier)**.** *Under the same ordinary native, compiler, and refutation calibration as the pinned scalar-carrier parent,*

$$\mathrm{ZFC} \vdash \Big(\forall x \in \mathbb{Z}^{12}\ P(x) \neq 0\Big) \longleftrightarrow \mathrm{Con(ZFC)}.$$

The new arithmetic equivalence is proved below and is formalizable in ZFC. This statement does not claim a new encoded object-ZFC derivation or proof-assistant endpoint for the entire inherited calibration.

Soundness. The fixed parity and residue choices give, unconditionally,

$$|q| \geq 16, \quad |X| \geq 64, \quad |Y| \geq 4096, \quad H \neq 0, \quad \Delta = (a+2)^2 - 1 > 1.$$

Also $E \geq 1$, since $\ell(\ell+1) \geq 0$ for integer ℓ. If $U = 0$, then $Q = 0$; otherwise $Q \geq \Delta^2 > 1$. Thus $\delta \neq 0$ for every integer assignment.

Write $\pi(x) = (x^2+3)(x^3-19)$. If $U = 0$, then $S = 0$ and the final polynomial becomes A_c^8. But $A_c = H^5\pi(T + K/H) \neq 0$: the quadratic has no real zero and the cubic has no rational zero. Hence a zero of P has $U \neq 0$, so s, r, c are nonzero and $|U| \geq c^2$. We obtain $D_f > |c| + 1$. Since b is integral, either $b = 0$ and $|V| = |c| \geq 1$, or $|V| \geq D_f - |c| > 1$. Consequently $C_y > 0$ and $Q > 1$ before any guard or projection is assumed.

Divide a vanishing conjugate factor of (21) by S. If $D_g < 0$, its summand is the only nonzero imaginary term, an impossibility. Thus $D_g \geq 0$. Since $s \neq 0$, the inequality $\Pi \leq 0$ would force $D_g \leq -3$; hence $\Pi > 0$. This implies both $G > 0$ and $\eta\zeta > 0$. Indeed their product is positive; if they were both negative, their absolute values $A, B > 0$ would give $\Pi = AB - (t^2 + A + B + 1)^2 < 0$.

Modulo 3, $X = -1$ and $a = H = \Delta = 0$. The value $\ell = XY^2$ is 0 or -1, so $E = D_f = 1$ and $\delta = C_y = -1$ modulo 3. The three normalized roots have squared 3-adic orders

$$\sqrt{C_y/\delta} : 0, \qquad 3\sqrt{ED_f} : 2, \qquad 9\sqrt{D_g} : 4 + v_3(D_g).$$

Omit a zero guard term. In a rational sum of square roots of positive rationals, each nontrivial square class must cancel internally. Within such a class the ratios are rational, and the displayed valuations give a unique term of minimum 3-adic order, making cancellation impossible. Therefore all three roots are rational.

The integer ED_f is thus a square. Since $D_f = 1 + \Delta U^2 E$, we have $\gcd(E, D_f) = 1$; positivity and unique factorization give

$$E = \tau^2, \qquad D_f = f^2, \qquad I = U\tau, \qquad \tau, f \in \mathbb{Z}_{>0}.$$

This includes $E = 1$. We have $c^2 \mid I$, $I \neq 0$, $f^2 = 1 + \Delta I^2$, and $Q = (\Delta I)^2$. The rational auxiliary root y satisfies

$$QV^2 - (Q-1)y^2 = 1. \tag{22}$$

The adjacent-coefficient descent from the retained parent makes y integral. Explicitly, if $|y| = a/b$ in lowest terms then $b^2 \mid Q-1$; the norm-preserving map

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV$$

preserves the reduced denominator and strictly decreases $|V|$ until $|V| = 1$, where $|y| = 1$. Signs are reflected during the descent. The guard root is integral as well.

The vanishing factor now gives

$$S(n+V) - A_c = S(\pm 3\tau f \pm 9g \pm y),$$

so $S \mid A_c$ and $H^5 \mid A_c$. Expanding

$$\Phi_H(z) = z^5 + 3H^2z^3 - 19H^3z^2 - 57H^5,$$

if $p^e \| H$ and $v_p(z) = b < e$, its leading term has valuation $5b$ and every other term has larger valuation. Thus $H^5 \mid \Phi_H(z)$ implies $H \mid z$. Applying this to $z = HT + K$ recovers $H \mid K$, with no sign assumption on H.

The exact source coordinates are

$$r_{\mathrm{old}} = r\tau, \qquad T_{\mathrm{old}} = \tau T, \qquad I = s r_{\mathrm{old}} c^2. \tag{23}$$

They give

$$4IT_{\mathrm{old}} = 4UET, \qquad V = f^2(c - R + 4IT_{\mathrm{old}}) - c.$$

All strict guards, the first and strong norm squares, the auxiliary norm (22), $c^2 \mid I$, and $H \mid K$ are now exactly the hypotheses of the pinned signed scalar-carrier restoration theorem. That theorem permits arbitrary nonzero strong multipliers, including even ones.

For precision, it uses absolute Pell folding and Lucas multiplication to recover $|c| = \psi_{|a+2|}(R)$; the auxiliary residues and strict index bounds identify the index. Its whole-product estimate, using $H \mid K$, then yields $\mathrm{sign}(c)X = 2^R$ and restores the missing main root and generalized signed native kernel. These unchanged lemmas are retained in `integer12_saturated/parent/PROOF.md` and its pinned signed proof sources. The named native/refutation calibration therefore supplies a finite syntactic ZFC refutation. The claim is restoration of that kernel, not a pointwise substitution in the old scalar polynomial, whose carrier uses T_{old}. □

6.3 Completeness with a fixed strong witness

Start with the canonical native witness of a finite ZFC refutation. Its relevant conditions are

$$A_0 = a + 2 > 1 \text{ even}, \quad R \equiv 3 \pmod 4, \quad c = \psi_{A_0}(R) > 0, \quad c \equiv R \pmod 4, \quad E = \tau^2, \quad \tau > 0.$$

Choose t for the joint strict guard, then a positive s with $D_g = g^2$, as in the parent. When $\Pi = 1$ one can use $s = 1, g = 0$; otherwise ordinary Pell existence supplies the guard witness.

The full-residue interpolation theorem in the retained `integer12_saturated/parent/source/INTERPOLATION12.md`, Section 6, says that for *every* prescribed positive N, there is one fixed strong pair I, f with $4Nc^2 \mid I$, such that *every* residue class modulo that same I is attained by an auxiliary T_{old} satisfying the norm. Its strong pair is chosen before the residue request, and its squared-modulus period is $4mf$, where m is the strong Pell index.

Apply this theorem with $N = s\tau$, obtaining

$$4s\tau c^2 \mid I, \qquad U = I/\tau, \qquad r = I/(s\tau c^2).$$

Thus $U = src^2$ is integral. The canonical main projection gives $H \mid K$; put $\kappa = K/H$. The retained intersectivity theorem for $\pi(x) = (x^2+3)(x^3-19)$ provides a residue a_* modulo U with

$$\pi(a_* + \kappa) \equiv 0 \pmod{U}.$$

Now request the old auxiliary residue

$$T_{\mathrm{old}} \equiv \tau a_* \pmod{I}.$$

Since $I = \tau U$, this ensures $\tau \mid T_{\mathrm{old}}$ and, on setting $T = T_{\mathrm{old}}/\tau$, gives exactly $T \equiv a_* \pmod{U}$. The new auxiliary expression is therefore the required source expression, and

$$\frac{A_c}{S} = \frac{\pi(T+\kappa)}{U} \in \mathbb{Z}.$$

Finally choose

$$n = \frac{A_c}{S} + 3\tau f + 9g + y - V \in \mathbb{Z}.$$

This makes a conjugate factor vanish and proves completeness. No positivity of n, coprimality of τ and U, or change of strong pair after the residue choice is needed. The factor E in $4UET$ is essential to this exact quotient-coordinate argument.

6.4 Attained degree and reproducible cost

All fixed coefficients have degree zero. The relevant total degrees are

	U	E	Δ	D_f	δ	b	V	C_y	S	A_c
deg	4	34	12	54	66	39	93	252	34	45.

Put $z_0 = Sn - A_c$, of degree 45, and $\omega = f_s - f_c$. The defining relation $C_y = (\delta+1)V^2 - 1$ gives the exact cancellation

$$\omega = \delta z_0^2 + 2\delta SV z_0 - S^2(V^2-1).$$

Its unique top term is $-S_0^2V_0^2$, of degree 254; competing degrees are 156, 238, and 68. The four squared arguments have degrees

$$\deg f_a = 222, \qquad \deg f_b = 180, \qquad \deg f_c = \deg f_s = 320.$$

The identity

$$P_e = \omega^2 - 2(f_a+f_b)(2f_c+\omega) + (f_a-f_b)^2$$

has unique top term $-4(f_a)_0(f_c)_0$, of degree 542; every competitor has degree at most 508. The correction $64f_sf_cf_af_b$ has degree 1042, so the final degree is $2 \cdot 542 = 1084$, provided the top form does not vanish at the fixed tuple.

It does not. With subscript zero denoting highest homogeneous part,

$$\begin{aligned}
q_0 &= 6\beta j, & X_0 &= 12q_0w, & Y_0 &= 2q_0^3v,\\
a_0 &= X_0Y_0, & U_0 &= src^2, & E_0 &= h^2X_0^4Y_0^6,\\
S_0 &= 2^{10}a_0^5U_0, & \delta_0 &= a_0^4U_0^2E_0, & (D_f)_0 &= a_0^2U_0^2E_0,\\
V_0 &= 4a_0^2U_0^3E_0^2T.
\end{aligned}$$

The term $4UET$ uniquely dominates R in b, so these are single monomials. The entire degree-1084 homogeneous part is

$$P_0 = 2^{1144}3^{760}\beta^{624}j^{624}w^{132}v^{164}h^{32}s^{32}r^{32}c^{64}T^4 \neq 0. \tag{24}$$

The exponents sum to 1084 and the actual fixed β is positive. Thus the degree is proved for the complete polynomial, independently of finite line tests.

The literal operation count is 75 multiplications, 25 additions, and 23 subtractions: 123 in total, including scalar multiplications. Every arithmetic node is live. The 13-operation final norm follows a 110-operation prefix and carrier. An included integral control has the same operation count and degree 1280; it is dominated by this candidate.

The independent standard-library checker imports no producer. It verifies every prefix by free-ring identities, all 35 coefficients of the eight-conjugate norm, the centered cancellation and restoration identities, exact leading induction, complete coefficient bytes and recipe, and operation counts. Normal and optimized runs pass; corrupted auxiliary, strong, guard, carrier, and final expressions are rejected. These computations certify the literal polynomial and its algebraic metrics. The signed semantic argument and interpolation proof are ordinary mathematical proofs using the explicitly retained source interfaces, independently reviewed in the accompanying audit.

7 Proof dependencies, reproducibility, and limitations

7.1 The inherited semantic boundary

The new positive proofs reuse the following ordinary theorems from the pinned construction. Their exact statements and source proofs are retained under `source/proofs/`:

1. Main/auxiliary Pell folding recovers $c = \psi_A(p)$ from the strong equation, the auxiliary equation, and their retained congruences.
2. The strong rank and signed auxiliary-index lemmas recover $p = R$ once the transport argument has supplied the correct rank range.
3. The zero-offset Pell-ratio estimate, whole-product projection, fixed-minus parity, and reconstruction of the unscaled source apply to the recovered outer data. In the direct variant, delayed q-divisibility is part of this interface.
4. The complete native/controller theorem, together with the K215 syntactic-refutation calibration and the coherent BFS state/color transfer, identifies source solutions with finite ZFC refutations.

The fourth item is a substantive inherited theorem. Reconstructing the coefficient bytes or running a finite machine prefix would not prove it. This packet does not describe those finite computations as such a proof. The new arithmetic arguments are uniform for the specified port box; the final fixed tuple is the *whole* coherent BFS tuple from the complete source page. Its K_0 is not equated with the different original compact44 K_0. No individual port is substituted between tuples.

The twelve-integer packets have their own six-port source34 tuple, `mixed-parity-unrounded-1064921`. The parent and dependencies are retained separately under `integer12_saturated/parent/` and `integer12/parent/`. The two coefficient families are not mixed. All ordinary proofs use only finite algebra, integer arithmetic, induction, and the previously arithmetized source calibration. The remaining end-to-end proof-assistant work is explicitly outside this contribution.

7.2 What the executable checks establish

- Every literal gate, input, fixed port, and live operation count is checked; deterministic rebuilds reproduce the exported JSON files.
- The complete generic two-root norm has six terms after expansion; its equality with the four signed factors is verified exactly. The intermediate and saturated integer three-root norms, and the earlier twelve-integer sixteen-factor norm, are checked independently as well.
- The anti-cancellation congruence and shared-product identities are checked over free polynomial rings, independently of sample values.
- The complete four compact44 coefficients are reconstructed from the printed 436-character seed and all 943 tiles, then compared against every supplied coefficient byte. The twelve-integer coefficient recipe is replayed separately.
- Exact univariate polynomial replays corroborate all four saturated degree certificates. The global upper bounds and nonzero highest homogeneous forms are proved symbolically in the article; they are not inferred from those one-dimensional checks.
- Independently transcribed complete formulas are bound to every gate of the four positive saturated graphs and the twelve-integer saturated graphs by exact identities over free local polynomial rings. Modular evaluations with the actual fixed coefficients provide additional transcription controls. These checks do not prove integer unsolvability.

From the extracted archive root, the main finite replay commands are

```
python3 -B build.py --check
python3 -B verify.py
python3 -B -O verify.py
python3 -B integer12_saturated/build.py --check
python3 -B integer12_saturated/verify.py
python3 -B -O integer12_saturated/verify.py
python3 -B integer12/build.py --check
python3 -B integer12/verify.py --output /tmp/integer12-check.json
python3 -B alternative/relational_kernel.py
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

Only the Python standard library is needed for the arithmetic checks. LaTeX is needed only to rebuild the PDF. No network access is needed after extraction. Verification receipts and provenance hashes are included, together with the complete sources.

8 Further research questions

1. **Can saturation be applied to more square constraints without increasing witness-reconstruction cost?** Coprimality alone is not enough: every absorbed factor must be supplied by a compatible canonical witness. The Pell-period argument here solves that problem for the first and strong ordinates.
2. **Can the upper center be approximated more accurately by a cheap integer polynomial?** The sharp center AUE leaves defect $(U^2E^2 - E)/(AUE + \sqrt{ED_f})$. Removing its highest

homogeneous part could lower degree further, but extra denominators and their exact integer recovery must be counted.

3. **Can a nine-positive form use the same two-root interface?** Eliminating an additional interpolation coordinate requires a proved simultaneous Pell-divisibility construction. Setting a witness to 1, or sharing two independent ordinates without that theorem, does not establish completeness.

4. **Can a source based directly on equational proofs beat the native history compiler?** The local kernel in the final appendix is explicit and small. A fully printed finite axiom tuple, a conservative calibration, and a counted unbounded-history compiler are still needed before it becomes a competing consistency equation.

5. **Which ordinary calibration links should be formalized next?** A useful next target is the new product-square recovery and the finite Pell-period completeness map, followed by composition with the existing native/internal-arithmetic work. Rechecking only finite evaluations would not discharge that task.

A The complete fixed coefficient recipe

The following finite recipe is copied from the complete, calibrated BFS44 page at the pinned snapshot. Every sum, index, floor, alphabet symbol, and tile in this appendix defines a fixed integer before the polynomial is evaluated. The independent reconstruction in `coefficients/recipe_compact44.py` reproduces every coefficient byte.

Four integers

Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1+2^{51})G_{943}(x) - 2^{10}\sum_{t\in\mathcal{T}} v(t)x^{\varrho(t)} \\ &\quad + \sum_{(b,e,m)\in\mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10,1023,8),(18,1013,9),(27,1003,8), \\ &\quad (35,983,9),(44,1031,1),(45,1022,1),(46,1011,1), \\ &\quad (47,992,1),(48,1095,1),(49,2195,1),(50,3296,1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 tiles and their ranks

Use the actions u_j, v_j, d_j decoded below. Let

$$h(q,b) = 2(q+1)+b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\},$$
$$\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\},$$
$$r_q = 6 + |\mathcal{R} \cap \{0,\dots,q-1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0,\dots,q-1\}|.$$

An interval ending at -1 is empty. The following union is $\mathcal{T}$; the index ranges in each row are fixed:

range	tile (w,n,e,s)
$j = 2q+b < 430$	$((1-d_j)l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$b = 0,1$	$(0, h(215,b), 0, h(215,b))$, $(0,b,0,b)$
$q \in \mathcal{R},\ b = 0,1$	$(r_q, b, 0, h(q,b))$
$q \in \mathcal{L},\ b = 0,1$	$(0, b, l_q, h(q,b))$
$t \in \{0,1,432,433\}$	$(0,t,0,438)$
	$(2,434,2,0)$, $(2,435,3,1)$
	$(3,436,4,h(u_0,0))$, $(4,436,4,0)$, $(5,0,0,0)$
$t = 434,435,436$	$(1,438,1,t)$
$a \in \{0,1,4\},\ b = 0,1,2$	$(a,437,b,437)$

Put $s_* = (2,435,3,1)$, $f_* = (5,0,0,0)$, and $\mathcal{T}' = \mathcal{T} \setminus \{s_*, f_*\}$. For $t = (w,n,e,s)$ set

$$\kappa(t) = 2^{30}w + 2^{20}n + 2^{10}e + s,$$
$$v(t) = w + 2^8 n + 2^{17}e + 2^{25}s + 127\,2^{34},$$
$$\varrho(s_*) = 0, \quad \varrho(f_*) = 1,$$
$$\varrho(t) = 2 + \sum_{z\in\mathcal{T}'} \left\lfloor \frac{2^{40} + \kappa(t) - \kappa(z) - 1}{2^{40}} \right\rfloor \quad (t \in \mathcal{T}').$$

A finite decoder for the 430 actions

Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$

The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

The archive uses little-endian unsigned coefficient bytes, compressed with gzip. The bit lengths of β, M_C, M_F, K_0 are respectively $590069, 590069, 590070, 592884$. Here the appendix symbol K is the polynomial port K_0; M_F already includes its prescribed shift by β. The accompanying payloads constitute one coherent tuple.

B Coordinate-shear controls over twelve integers

The following two earlier improvements are retained as simple algebraic controls. The saturated equation in Section 6 has a better operation count and degree than either. These controls use a different proof mechanism: invertible polynomial coordinate changes covering the entire integer domain, including exceptional values of the denominators used only in explanations.

The preceding positive-domain constructions and the following integer-domain constructions are separate points in the comparison. Here all twelve unknowns range over $\mathbb{Z}$. We give two polynomial coordinate changes of the pinned repository's current scalar-carrier equations. The transformations preserve the entire integer zero sets bijectively and improve their degree–cost tradeoffs. Centering is already used elsewhere in the repository; the contribution here is its application to the present scalar four-root carrier, with complete circuits and attained degree certificates.

B.1 Fixed coefficients and complete equations

The ordered unknowns are

$$(j, C, z, h, r, v, w, T, c, n, t, s) \in \mathbb{Z}^{12}.$$

Use the coherent `mixed-parity-unrounded-1064921` tuple, independently of the compact44 tuple used by the positive-domain constructions:

$$\beta = 2^{1064921} - 1, \qquad \kappa_1 = 2^{1113}, \qquad \kappa_2 = 2^{666104}.$$

The remaining fixed integers K_0, M_C, M_F are the exact source34 values in the included coefficient payloads. The complete finite recipe, literal machine seed, and all 943 tiles are included; the verifier rebuilds every coefficient and compares all bytes. In particular M_F is already shifted by β. The six coefficients are never mixed with another tuple.

Define the following polynomial abbreviations; none is an additional unknown:

$$\begin{aligned}
J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4,\\
X &= qW, & Y &= (2v + 1)q^3, & E &= XY,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H,\\
F &= (K_0 + W)C - z(q - 1),
\end{aligned}$$

$$\begin{aligned}
R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J,\\
k &= R + 1 + hE, \qquad \eta = c - Yk, \qquad \zeta = k - \eta,\\
G &= (\kappa_1 F)(\kappa_2 C) - (q - \kappa_1 F - \kappa_2 C)^2,\\
\Pi &= G\eta\zeta - (t^2 - G - \eta\zeta + 1)^2,\\
D_g &= (3\Pi - 4)s^2 + 1, \qquad I = src^2,\\
D_f &= 1 + \Delta I^2, \qquad Q = \Delta^2 I^2, \qquad \delta = Q - 1,\\
V &= D_f(c - R + 4IT) - c, \qquad C_y = QV^2 - 1,\\
\ell &= XY^2, \qquad D_t = 1 + \ell(\ell + 1)k^2,\\
K &= X(X + 2ac) - 1, \qquad S = IH^5, \qquad z_c = HT + K,\\
A_c &= (z_c^2 + 3H^2)(z_c^3 - 19H^3).
\end{aligned}$$

Set

$$r_a = 9S^2 D_f, \qquad r_b = 81S^2 D_t, \qquad r_e = 729S^2 D_g.$$

Let $\mathcal{F}_{16}(Z; u_1, u_2, u_3, u_4)$ be the product of $Z - \sum_{i=1}^{4} \epsilon_i \sqrt{u_i}$ over the sixteen choices $\epsilon \in \{-1, 1\}^4$. Sign invariance makes this an integer polynomial in its five arguments; every radicand has exponent at most

eight. Our two equations are

$$P_R = \delta^8 \mathcal{F}_{16}\left(S(n+V) - A_c; r_a, r_b, r_e, \frac{S^2 C_y}{\delta}\right) = 0, \tag{25}$$

$$P_I = \mathcal{F}_{16}\left(S(n+\delta V) - A_c; r_a, r_b, r_e, S^2 \delta C_y\right) = 0. \tag{26}$$

The first formula means its symbolic polynomial after denominator clearing, not numerical evaluation by division. Both literal JSON circuits contain only fixed integers, addition, subtraction, and multiplication.

For completeness, the following polynomial recursion removes every radical from the notation. For a center Z, put $d = \delta$ and $c_n = S^2 C_y$ in (25), or $d = 1$ and $c_n = S^2 \delta C_y$ in (26). Define

$$\begin{aligned}
t_0 &= Z^2 + r_e, & t_1 &= -2Z, & b_0 &= r_a + r_b,\\
q_0 &= d(t_0 - b_0) - c_n, & q_1 &= d t_1,\\
T_0 &= d^2 r_a r_b, & R_0 &= d c_n,\\
v_0 &= q_0^2 + r_e q_1^2 - 4(T_0 + R_0 b_0), & v_1 &= 2 q_0 q_1,\\
z_{64} &= 64 T_0 R_0,\\
n_0 &= v_0^2 + r_e v_1^2 - z_{64} t_0, & n_1 &= 2 v_0 v_1 - z_{64} t_1.
\end{aligned}$$

Then the polynomial output is $n_0^2 - r_e n_1^2$. The independent oracle checks all 495 coefficients against the sixteen-conjugate product.

B.2 Unconditional integer zero-set bijections

Write P_R^{old} and P_I^{old} for the pinned parent equations with the same prefix and center $Sn - A_c$. Every other abbreviation is independent of n. The maps

$$T_R:\ n_{\mathrm{old}} = n + V, \qquad T_I:\ n_{\mathrm{old}} = n + \delta V$$

fix eleven coordinates. Their inverse maps subtract V and δV, respectively. These are mutually inverse polynomial maps of $\mathbb{Z}^{12}$, without sign, nonzero, divisibility, or zero-equation hypotheses. Direct substitution gives the identities

$$P_R = P_R^{\mathrm{old}} \circ T_R, \qquad P_I = P_I^{\mathrm{old}} \circ T_I.$$

Consequently each pair has bijective complete integer zero sets, including all exceptional branches as formal polynomials. Composing these elementary identities with the unchanged ordinary parent calibration gives, in the same formalizability sense,

$$\mathrm{ZFC} \vdash \left(\forall x \in \mathbb{Z}^{12}\ P_R(x) \neq 0\right) \leftrightarrow \mathrm{Con}(\mathrm{ZFC}),$$

and the identical statement with P_I. This composition introduces no new native compiler claim. The retained parent proof and its dependencies remain part of the evidence; this section does not claim a new machine-checked object-ZFC derivation.

B.3 Exact degree certificates

Only quantified coordinates contribute to total degree. The prefix gives

$$\deg S = 34, \quad \deg V = 25, \quad \deg \delta = 32, \quad \deg r_a = 88, \quad \deg r_b = 102, \quad \deg r_e = 114, \quad \deg A_c = 45.$$

Put $z_0 = Sn - A_c$, of degree 45. For P_R, the relation $C_y = (\delta+1)V^2 - 1$ gives the exact cancellation

$$\delta(SV + z_0)^2 - S^2 C_y = \delta z_0^2 + 2\delta S V z_0 - S^2(V^2 - 1).$$

The right side has degree at most 136. Thus the norm register q_0 has unique highest part $\delta_0(r_e)_0$ of degree 146, while $(q_1)_0 = -2\delta_0 S_0 V_0$ has degree 91. Subsequent strict degree comparisons give

	q_0	q_1	T_0	R_0	v_0	v_1	n_0	n_1	P_R
deg	146	91	254	182	296	237	592	533	1184.

Here and below a subscript zero on a polynomial denotes its highest homogeneous component, independently of the temporary norm-register names. The full top form is

$$(P_R)_0 = 2^8 3^{24} S_0^{16} \delta_0^8 V_0^8 (D_g)_0^4.$$

For P_I, the analogous identity is

$$(S\delta V + z_0)^2 - S^2 \delta C_y = z_0^2 + 2S\delta V z_0 - S^2 \delta V^2 + S^2 \delta.$$

Its unique highest term is $-S_0^2 \delta_0 V_0^2$, of degree 150. The remaining induction gives

	q_0	q_1	T_0	R_0	v_0	v_1	n_0	n_1	P_I
deg	150	91	190	182	300	241	600	541	1200,

and $(P_I)_0 = S_0^{16} \delta_0^8 V_0^{16}$.

These bounds are attained on the actual fixed tuple, not merely on generic symbolic ports. Selecting the largest T power in each top form gives

$$[j^{624} w^{144} v^{160} s^{64} r^{56} c^{112} h^{16} T^8] P_R = 2^{1256} 3^{796} \beta^{624} > 0,$$
$$[j^{576} w^{144} v^{144} s^{80} r^{80} c^{160} T^{16}] P_I = 2^{1200} 3^{720} \beta^{576} > 0.$$

The exponent sums are 1184 and 1200. These are distinguished top-degree monomials, not assertions that the full homogeneous components contain only one term. In particular the top part of V contains $4I_0 T - R_0$, whose two summands both have degree five. The exact checker keeps the graded-lex leading term with T first and confirms every coefficient induction after the displayed cancellation.

B.4 Cost and what the computations certify

Under the same fully charged binary-operation convention as the parent, the literal counts are

Equation	Multiplications	Additions	Subtractions	Total	Degree
Old rational	93	26	26	145	1200
New P_R	93	27	26	146	1184
Old integral	89	26	26	141	1456
New P_I	90	27	26	143	1200

Thus P_I reaches the old rational degree with two fewer operations, while P_R supplies a lower-degree option within this historical comparison. The saturated 123/1084 equation of Section 6 improves both metrics of every row.

The standard-library verifier imports no new producer. It proves the whole-output substitution identities by a common expression DAG, checks both integer inverse maps, binds all prefix formulas by independent free-ring cuts, verifies every generic norm coefficient, and performs the exact leading induction. It also reconstructs all fixed coefficients from the finite recipe and checks the live operation counts. Normal and optimized Python runs pass, and corrupted center and norm signs are rejected. These checks establish exact polynomial algebra and transcription; the semantic conclusion uses the unbounded bijection proof and the named ordinary parent theorem.

C An alternative source: an explicit equational proof kernel

This appendix supplies a concrete local component of a different source encoding. It does not supply a new consistency equation. The broad route through finitely axiomatized class theory and relation algebra is prior work: Carl and Moroz used finite NBG in their Diophantine provability construction, and Formisano, Gavazzi, and Omodeo subsequently proposed a Tarski–Givant relational version. The latter paper does not print its proper-axiom tuple; its Section 3.3 leaves those formulas outside its scope. Consequently it cannot, by citation alone, calibrate a concrete new polynomial to ZFC. Our contribution here is an independently derived, completely explicit local inference polynomial and an executable finite proof checker for a supplied finite collection of equation schemata. No priority claim is made for the local algebraic identity.

C.1 A total coding of terms

Work over $\mathbb{N} = \{0, 1, 2, \ldots\}$. Let the abstract term language have two constants ϵ, ι, two unary constructors N, V, and two binary constructors U, M. These will eventually stand for membership, identity, complement, converse, union, and relation composition, respectively. The coding theorem itself is purely syntactic and does not require that interpretation. Set

$$\begin{aligned} [\epsilon] &= 0, & [\iota] &= 1, \\ [N(A)] &= 4[A] + 2, & [V(A)] &= 4[A] + 3, \\ B(a, z) &= 2(a + z)(a + z + 1) + 4z + 4, & & \\ [U(A, Z)] &= B([A], [Z]), & [M(A, Z)] &= B([A], [Z]) + 1. \end{aligned}$$

Every displayed constructor is a polynomial with integer coefficients. In particular there is no division in the code of a compound term. This is the polynomial form of the numbering in Section 3.1 of Formisano–Gavazzi–Omodeo; the complete bijection proof is included here to make the component independently checkable.

Theorem A.1 (total coding). The coding map is a bijection from these finite terms onto $\mathbb{N}$.

Proof. The Cantor function

$$C(a, z) = \frac{(a + z)(a + z + 1)}{2} + z$$

is a bijection $\mathbb{N}^2 \longrightarrow \mathbb{N}$. Indeed, for a given g, the unique s satisfying $s(s + 1)/2 \leq g < (s + 1)(s + 2)/2$ gives $z = g - s(s + 1)/2$ and $a = s - z$. Then $0 \leq z \leq s$, so these are natural numbers, and the formulas also prove uniqueness. Moreover $C(a, z) \geq \max(a, z)$.

The two constants occupy $0, 1$. Codes congruent to 2 and 3 modulo 4 have the unique forms $4a + 2$ and $4a + 3$, respectively. The remaining numbers are $4(g + 1)$ and $4(g + 1) + 1$ with $g \geq 0$; the unique inverse of C identifies their two children. Since $B = 4(C + 1)$, these are precisely the two binary constructor cases. In every case the child codes are strictly smaller than the parent code. Strong induction on a natural number therefore constructs one finite term having that code; the disjoint residue cases and the uniqueness of the Cantor inverse prove that this term is unique. This also proves injectivity of the original coding map. All inductions and integer computations are available in ZFC. □

Thus a natural-number witness never needs a separate certificate that it denotes a valid term. Equalities are stored as two coordinates (a, b), representing the equation between the uniquely decoded terms. This avoids packing and immediately unpacking an equality as a further Cantor pair.

C.2 The seven-variable local polynomial

Let (a, b) and (c, d) be two already proved equations and let (u, v) be a proposed conclusion. Introduce only one additional natural witness z. Define

$$\begin{aligned} T &= (a-c)^2 + (u-b)^2 + (v-d)^2, \\ q &= u - 4a - 2, \\ A &= \big(q(q-1)\big)^2 + \big(v - u - 4(b-a)\big)^2, \\ r &= u - B(a, z), \\ D &= \big(r(r-1)\big)^2 + \big(v - u - 2(b-a)(a+b+2z+1)\big)^2, \\ K(u, v, a, b, c, d, z) &= TAD. \end{aligned}$$

There are seven variables in K, not seven witnesses for the whole provability problem: six coordinates are inputs and outputs of a single local inference. The history encoding is a separate cost.

Theorem A.2 (exact local inference). For natural values of its seven arguments, $K = 0$ holds if and only if at least one of the following five inference cases holds:

1. $a = c$, $u = b$, and $v = d$ (common-left transitivity);
2. $u = 4a + 2$ and $v = 4b + 2$ (complement congruence);
3. $u = 4a + 3$ and $v = 4b + 3$ (converse congruence);
4. $u = B(a, z)$ and $v = B(b, z)$ (union in the right argument);
5. $u = B(a, z) + 1$ and $v = B(b, z) + 1$ (composition in the right argument).

Here "in the right argument" means adjoining the same right operand z to both members of a proved equality; it does not mean changing that right operand.

Proof. The integer ring has no zero divisors, so $K = 0$ is equivalent to the vanishing of one factor. Each factor is a sum of squares of integers. For T, this gives exactly case 1. For A, it gives $q(q-1) = 0$ and $v - u = 4(b-a)$. Hence q is 0 or 1 and the same unary tag is used on both sides, proving cases 2 and 3.

For D, it gives $r \in \{0, 1\}$ and the stated difference equation. Direct expansion gives the polynomial identity

$$B(b, z) - B(a, z) = 2(b-a)(a+b+2z+1).$$

Thus $u = B(a, z) + r$ and $v = B(b, z) + r$, proving cases 4 and 5. Every listed case conversely annihilates its indicated factor. The argument establishing this union of arithmetic cases works over all integers; nonnegativity is used only for the term interpretation from Theorem A.1. □

The two adjacent constructor tags are therefore selected without introducing a Boolean variable. The finite-difference identity also avoids calculating $B(a, z)$ and $B(b, z)$ independently. The included straight-line circuit has 41 live binary additions, subtractions, and multiplications; a square counts as one multiplication, and its constants are $1, 2, 4$. The circuit is an evaluator for this local polynomial, not an operation count for a completed consistency equation.

Exact degree. The three factors have degrees $2, 4, 8$, respectively, so $\deg K \leq 14$. Specializing $a = t$ and all other inputs to zero gives leading terms t^2, $256t^4$, and $16t^8$. Therefore the resulting univariate polynomial has leading term $4096t^{14}$, proving $\deg K = 14$.

C.3 Finite proof checking and the remaining obligations

Fix a finite list of equation schemata. Each side is a finite term template over the six constructors, with a finite number of schematic variables. Supplying natural values for those variables produces an exact instance by the constructor polynomials. The code contains no axiomhood oracle: the list of templates is literal input data.

A proof line records its conclusion, and either (i) a schema index and its instantiation values, or (ii) two strictly earlier line indices and the witness z. A line of kind (ii) is accepted exactly when $K = 0$ for its two parent equations. One-parent congruence steps repeat the same parent index. The first line must be a schema instance. This explicit requirement excludes an unjustified cyclic first line.

Theorem A.3 (finite-checker adequacy). For the supplied schema list, the checker accepts a proof of a target equation exactly when that equation has a finite derivation from those schemata using the five inference cases in Theorem A.2.

Proof. If the checker accepts, induction over its finite list proves each line. At a schema line this follows by exact instantiation. At an inference line both parents have smaller indices, so the induction hypothesis and Theorem A.2 give its conclusion. Conversely, list the nodes of a finite derivation tree in a children-before-parent order. Record the two earlier indices for a transitivity step, duplicate the earlier index for a congruence step, and use the code of its right operand for a binary congruence. For the other steps $z = 0$ suffices. Theorem A.2 makes the kernel vanish at each inference, and exact constructor evaluation verifies every schema instance. □

This theorem concerns the stated finite equational calculus, without a claim that an arbitrary schema list expresses set theory. A finished replacement for the existing Con(ZFC) equation still requires all of:

1. A completely printed finite relational class-theory axiom tuple, including the logical relation-algebra schemata and a precise target equation expressing collapse.
2. Proof transformations, formalizable in ZFC, establishing that this target is derivable exactly when the chosen ZFC calculus is inconsistent. In particular the quasi-projection assumptions required for the Tarski–Givant translation must be proved or included correctly.
3. An explicit polynomial compiler for arbitrarily long accepted finite histories, with every bound, chronology condition, coefficient, and witness accounted for.
4. Complete global variable, degree, coefficient, and evaluator counts after that compiler is instantiated.

None of these global costs can be inferred from the seven local coordinates or 41 local operations. The component provides a distinct testable source direction, while leaving the established consistency equation unchanged.

C.4 An explicit combined row polynomial

There is also a direct polynomial form of schema admission. Suppose the jth supplied schema has at most m schematic variables. Evaluating its left and right templates using the constructor polynomials gives fixed polynomials $L_j(\mathbf{x}), R_j(\mathbf{x}) \in \mathbb{Z}[x_1, \ldots, x_m]$; unused coordinates are simply ignored. Define

$$H(u, v, \mathbf{x}) = \prod_{j=1}^{s} ((u - L_j(\mathbf{x}))^2 + (v - R_j(\mathbf{x}))^2), \qquad F = K(u, v, a, b, c, d, z)H(u, v, \mathbf{x}).$$

The empty product is 1. Over natural inputs, $F = 0$ means exactly that the output is a listed schema instance or follows from the two parents by one of the five inference cases. The same m schema coordinates may be shared between all s factors: a zero product requires only one schema factor to vanish. Conversely witnesses for that chosen schema can be padded with zeros. This observation does not share witnesses between multiple simultaneously required premises.

If the chosen logical basis includes reflexivity, histories can start with the fixed justified row $(0, 0)$. Every later row can then carry two strictly earlier parent indices, including axiom rows, which use the initial row as harmless dummy parents. Using $F = 0$ for those later rows avoids a separate Boolean axiom/inference tag. The first row is fixed and verified; it is never admitted merely by a self-referential inference. Induction gives the same soundness and completeness proof as above. This combined row polynomial still has an unbounded-history compilation obligation. In particular the variable count $7 + m$ is a local row count, not a variable bound for a Con(ZFC) equation.

Reproduction. Run `python3 relational_kernel.py`. The included finite checks cover 10,000 complete code roundtrips, 10,000 Cantor inverse roundtrips, 2,048 binary constructor cases, 20,000 seeded arithmetic comparisons, 8,640 positive inference cases, and valid/invalid finite proofs. These checks supplement the universal proofs above; they do not establish the missing NBG calibration or global history compilation.

Primary references. M. Carl and B. Z. Moroz, *On a Diophantine representation of the predicate of provability*, MPI preprint 2006/129, https://webdoc.sub.gwdg.de/ebook/serien/e/mpi_mathematik/2006/129.pdf. A. Formisano, I. Gavazzi, and E. G. Omodeo, *Diophantine Modeling of Provability in Algebraic Logic*, CILC 2025, https://ceur-ws.org/Vol-4003/paper13.pdf.

References

[1] V. Reshetnikov, *ProveIt: DiophantineConsistency*, snapshot `84e16e43092614bae262ca59651cb9d77e31a255`, inspected 24 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/84e16e43092614bae262ca59651cb9d77e31a255/Computability/DiophantineConsistency.

[2] ProveIt, *Ten strictly positive integers, 99 operations, degree 368*, `candidates/ten-positive-zero-offset`, at [1]. The complete ordinary proof and literal parent are retained in this packet.

[3] ProveIt, *Combined direct/zero-offset105/340*, `candidates/ten-positive-direct-zero-offset`, at [1].

[4] ProveIt source61, *Congruence carriers*, `con-zfc-diophantine-forms/61-congruence-carriers`; its rank, index, and auxiliary completeness interfaces are restated in the retained `ten-positive-scaled-norm.md` ordinary proof.

[5] ProveIt, *Twelve unrestricted integers by a scalar carrier*, `candidates/scalar-carrier12`, at [1]. The complete ordinary proof, signed-kernel interface, arbitrary-residue interpolation theorem, and literal parents are retained in this packet.

[6] ProveIt incoming submission, *ConZFC Dyadic Pell Reductions, 23 October 2026*, `dyadic10/PROOF.md`. Earlier distributed transport numerator and separate upper-root centering; full archive at the pinned incoming snapshot.

[7] M. Carl and B. Z. Moroz, *On a Diophantine representation of the predicate of provability*, Max Planck Institute preprint 2006/129. https://webdoc.sub.gwdg.de/ebook/serien/e/mpi_mathematik/2006/129.pdf.

[8] A. Formisano, I. Gavazzi, and E. G. Omodeo, *Diophantine Modeling of Provability in Algebraic Logic*, CILC 2025, CEUR Workshop Proceedings 4003, paper13. https://ceur-ws.org/Vol-4003/paper13.pdf.